

Optimal First- and Second-Price Auctions Under Dual Risk Aversion*

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Abstract

Among standard auction formats, the reserve-only first-price auction (FPA) maximizes the seller’s expected payoff when buyers are risk averse dual utility (DU) maximizers. We present an extension of this result for risk averse sellers. In particular, we show that strongly risk averse sellers (not necessarily DU maximizers) faced with weakly risk averse DU buyers prefer the optimal reserve-only FPA to the optimal reserve-only second-price auction (SPA), and do not benefit from entry fees under either format. In addition, we provide several new comparative statics for seller-optimal reserve prices under dual utility.

Keywords: auction design; risk aversion; dual utility; reserve prices; entry fees; probability weighting.

JEL: D44, D81, D82

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1 Introduction

A central goal of much auctions research is to provide clear and practical guidance to real-world sellers. In reality, sellers (like buyers) are frequently risk averse, or at least subject to constraints that give rise to apparent risk aversion (e.g., Froot et al., 1993). Yet many commonly held intuitions about auction design presume that sellers are risk neutral.

In this paper we compare seller-optimal first- and second-price auctions (FPA/SPA) when sellers are restricted to the use of reserve prices and entry fees. Such mechanisms arguably account for the bulk of real-world auctions of single, indivisible goods. In contrast to most prior work on risk aversion in auctions, we assume that agents are equipped with dual utility (DU) preferences (Yaari 1987). Within this framework, we show that strongly risk averse sellers faced with weakly risk averse buyers prefer a reserve-only FPA and do not benefit from entry fees in either the FPA or the SPA. These results, among others contained in the paper, are significant both insofar as they inform practical auction design and address long-standing questions regarding the value of entry fees when sellers are risk averse.

The implications of buyer risk aversion for auction design under classical expected utility (EU) are the subject of a substantial literature spanning economic theory, behavioral economics, and industrial organization.¹ The implications of seller risk aversion in such settings are less fully understood. In an early contribution, Waehrer et al. (1998) consider the case of an EU risk averse seller faced with risk neutral buyers. The seller prefers the FPA, because the induced distribution of FPA revenue is a mean preserving contraction of the distribution of SPA revenue. Optimal reserve prices fall below the classical level because EU risk averse sellers discount payoffs from successful sales. Hu et al. (2010) extend these results to the case in which buyers are also EU risk averse, a challenging task given the nonlinearity inherent to the EU model. Yet as Hu et al. (2010) acknowledge, it is not obvious whether a risk averse seller might prefer to screen buyers using reserve prices, entry fees, or a combination of the two: while Waehrer et al. (1998) show that entry fees are unhelpful when buyers are risk neutral, Maskin and Riley (1984) suggest that entry fees can increase revenue when buyers are risk averse. To our knowledge, clear guidance on this question remains scarce.²

Risk aversion in the strong sense is generally understood to describe preferences that respect second order stochastic dominance (Rothschild and Stiglitz, 1970).³ The DU model is one of many non-expected utility (NEU) models that generate strong risk aversion. In the DU model this is a consequence of non-linear probability weighting: agents that sufficiently discount the probability of high payoff states prefer mean preserving contractions. This

¹For a recent survey, see Vasserman and Watt (2021).

²Chakraborty (2019) shows that entry fees may be necessary for risk neutral sellers if buyers are EU maximizers with exponential utility. We consider risk averse sellers and buyers with general DU preferences.

³Strong risk aversion implies weak risk aversion, which refers to preferences for which an agent prefers the expectation of a lottery to the lottery itself. Weak risk aversion is all we will require of buyers.

contrasts with the classical EU model, where risk aversion is implied by the diminishing marginal utility of wealth. Relative to the EU model, the DU framework preserves the linearity of buyer and seller payoffs from the risk neutral model, enhancing tractability.

We begin by characterizing symmetric pure strategy equilibria in first- and second-price auctions in which buyers have general probability weighting functions. We focus on the classical setting of symmetric and independent private values. Notably, both auction formats admit closed-form equilibrium bidding strategies. In contrast to Volij (2002, 2025), our FPA characterization is based on a direct approach involving familiar differential equation techniques for first-price auctions rather than general payoff equivalence results.

Our main contribution is to show that, for the FPA and SPA, strongly risk averse sellers prefer reserve-only screening mechanisms. We do so by showing that the distribution of seller payoffs induced by a positive entry fee is second order stochastically dominated by the distribution induced by a reserve-only mechanism that screens the same potential buyers. This result does not depend on the behavioral foundation of the seller’s risk preference, but applies generally to any seller preferences that respect second order stochastic dominance (including EU risk aversion) provided that buyers are weakly risk averse DU maximizers.⁴

This result implies that we can restrict attention to reserve-only auctions. Holding the reserve price fixed, we show that the distribution of seller payoffs induced by the FPA second order stochastically dominates the distribution induced by the SPA when buyers are weakly risk averse DU maximizers. Hence, strongly risk averse sellers (whether DU, EU, or otherwise) prefer the FPA to the SPA for a fixed reserve. In turn, this implies that a strongly risk averse seller must prefer the optimal reserve-only FPA to the optimal reserve-only SPA. To illustrate how these may differ, we directly characterize seller-optimal reserve prices when the seller is a strongly risk averse DU maximizer. Our results are broadly similar to those of Hu et al. (2010) for the case of EU agents despite significant differences in behavioral foundations. In particular, we show that seller risk aversion reduces optimal reserve prices in both the FPA and SPA, with the former being strictly lower under a single-crossing condition.⁵ Increasing buyer risk aversion likewise reduces optimal reserves in the FPA.

There are several possible motivations for adopting the DU model apart from analytical convenience. For auctions in which agents are profit-maximizing firms or government entities (encompassing many of the most important real-world applications such as procurement and mineral rights leasing), non-linear von Neumann-Morgenstern utility in the EU model can be difficult to justify a priori. Once adopted, it implies that optimal auction design parameters

⁴As risk neutral buyers can be re-interpreted as weakly risk averse DU maximizers with a trivial distortion function, we therefore generalize Waehrer et al (1998)’s earlier entry fee result.

⁵More precisely, we show that strict risk aversion reduces optimal reserve prices below the familiar Myerson (1981) optimal reserve. To compare seller preferences represented by arbitrary convex distortions, we state a more general ranking result valid under single-crossing restrictions discussed in the text below.

are generally sensitive to agents’ initial wealth levels, which are rarely observed. These shortcomings limit the usefulness of EU, and non-linear payoff weighting more generally, for practical auction design (e.g., reserve price calibration). The DU model circumvents these challenges while offering a tighter connection to a variety of real-world risk measurement techniques that ultimately derive from non-linear probability weighting, such as the well-known expected shortfall or Conditional Value-at-Risk (see, e.g., Sarykalin et al., 2008).

Gershkov et al. (2020) show that for a risk neutral seller and DU risk averse buyers, the FPA with optimal reserve and zero entry fees is optimal within a large class of mechanisms that encompasses the FPA and SPA.⁶ We provide an extension for the case of risk averse sellers that addresses the two auction formats most commonly used in practice, in the spirit of Waehrer et al. (1998). As DU risk aversion encompasses risk neutrality, the paper also presents the opportunity to enrich the original risk neutral reserve price characterization from Gershkov et al. (2020). In particular, we aim to aid sellers and other practitioners by providing verifiable sufficient conditions under which seller-optimal reserve prices are unique, and by exploring how such prices vary with buyer and seller risk aversion.

We contribute most directly to the literature on auction design in environments with NEU preferences. Karni and Safra (1989) and Neilson (1994) studied the SPA under general NEU preferences. Che and Gale (2006) show that in the absence of screening, risk neutral sellers prefer the FPA to the SPA for a class of NEU buyer preferences that encompasses DU. More recently, Gershkov et al. (2022) show that neither the FPA nor the SPA is optimal in general for a risk neutral seller facing DU risk averse buyers: instead, the seller-optimal auction involves payments to potential buyers (i.e., negative entry fees).⁷ The reserve-only FPA that we characterize is therefore suboptimal within a broader class of auction mechanisms, though the optimal auction is likely to involve features rarely implemented in practice.

The rest of the paper proceeds as follows. Section 2 introduces the model and analyzes the bidding equilibrium. Section 3 considers optimal entry fees and reserve prices for risk neutral sellers. Section 4 considers risk averse sellers. Section 5 concludes.

2 Setting

This section introduces the model and describes the equilibrium.

⁶We refer here to the working paper version of Gershkov et al. (2022). The published version of the paper, which deals with a larger class of mechanisms (discussed below), omits the referenced results.

⁷The conclusions of Gershkov et al. (2022) differ qualitatively from the EU case studied in Maskin and Riley (1984), illustrating that the DU and EU models generally imply different optimal auction designs.

2.1 Model setup

We consider an auction for a single indivisible good. There are $N \geq 2$ symmetric potential buyers. Each buyer i has a private value v_i . We assume that the private values v_i are drawn independently from a distribution function $F(\cdot)$ with density $f(\cdot) > 0$ supported on $[\underline{v}, \bar{v}]$ for $\underline{v} \geq 0$. Let $v_0 \in (\underline{v}, \bar{v})$ be the seller's value. Throughout the paper, we assume the virtual value function $J(v) = v - \frac{1-F(v)}{f(v)}$ is strictly increasing. This assumption is commonly found in the literature and ensures the uniqueness of the optimal reserve under risk neutrality.

The seller is permitted to screen buyers using entry fees and reserve prices. Potential buyers first learn their valuations and then make simultaneous participation and bidding decisions. All potential buyers who choose to participate incur the entry fee, regardless of ultimate success. The realized number of participants $K \leq N$ is not observed by bidders.

Preferences In this paper, we adopt the dual utility model (DU) of Yaari (1987). Agents are equipped with probability weighting functions $g \in \mathcal{G}$, where $\mathcal{G}$ denotes the set of continuously differentiable and strictly increasing cumulative distribution functions (cdfs) on $[0, 1]$ that satisfy $g(0) = 0$ and $g(1) = 1$. Agents are *weakly risk averse* if $g(x)/x$ is weakly increasing on $(0, 1]$. In this case we say that g is star-shaped.⁸ We let $\mathcal{G}^* \subset \mathcal{G}$ denote the set of star-shaped distortions. Agents are *strongly risk averse* if $g(\cdot)$ is convex. All convex distortions are star-shaped, and hence we have $\mathcal{G}^{cvx} \subset \mathcal{G}^* \subset \mathcal{G}$ where $\mathcal{G}^{cvx}$ is the set of convex distortions. Weakly risk averse agents prefer a sure payment to a random variable with the same expectation. Strongly risk averse agents prefer mean preserving contractions.⁹

Let x be a random variable with bounded support $[\underline{x}, \bar{x}]$, potentially having mass points. Let H be the cdf of x . An agent's certainty equivalent of holding x (a lottery) is

$$U_g(x) = \underline{x} + \int_{\underline{x}}^{\bar{x}} g(1 - H(s)) ds = - \int_{\underline{x}}^{\bar{x}} s dg(1 - H(s)). \quad (1)$$

The first equality highlights the contrast between DU and EU preferences. In particular, DU agents' preferences are generally independent of initial wealth and invariant to the scale of the payoff distribution. In the EU model, the first property is satisfied only in the special case of constant absolute risk aversion (CARA) and the second only in the special case of constant relative risk aversion (CRRA); these properties are jointly satisfied only by risk neutrality. The second equality clarifies the connection to the classical risk neutrality: when g is the identity function, $dg(1 - H(s)) = -dH(s)$, and $U_g(x)$ is the expectation of x .

Common parametric examples of the distortion function g include the power distortion

⁸Because $g(1) = 1$, star-shapedness implies that $g(x) \leq x$ for all $x \in [0, 1]$.

⁹See, e.g., Armantier and Treich (2009). That convex g induces preferences that respect second order stochastic dominance is also implied by Chew et al. (1987) Corollary 2 (specialized to linear v^*).

$g(x) = x^\gamma$ for $\gamma > 0$ and the so-called Wang distortion $g(x) = \Phi(\Phi^{-1}(x) + \lambda)$ for $\lambda \in \mathbb{R}$, where Φ is the standard normal cdf. We discuss the first in our examples below.

In the equilibria that we study, buyers maximize their certainty equivalent when determining their participation and bidding strategies, and the seller maximizes her certainty equivalent when choosing the auction format and fixing any screening parameters.

2.2 Equilibrium analysis

Before analyzing the seller's problem, we derive symmetric increasing equilibrium bidding strategies under SPA and FPA. We first develop our analysis for the case of zero entry fees and a potentially binding reserve: $e = 0$ and $r \in [\underline{v}, \bar{v}]$. We assume that all buyers are equipped with the same symmetric probability weighting function $g \in \mathcal{G}$.¹⁰

As in the EU model and the classical risk neutral model, truthful bidding is a dominant strategy in the SPA: no buyer wishes to lose at a price below his true value, nor to win at a price above it. We restrict attention to the symmetric equilibrium with truthful bidding.

The FPA requires a more careful analysis. Let $b(\cdot)$ denote the equilibrium bidding strategy in a symmetric monotone pure-strategy equilibrium. It is clear that we must have $b(r) = r$, while only buyers with a value higher than r submit a bid. Thus, it remains to characterize $b(\cdot)$ over $(r, \bar{v}]$. This can be done directly by invoking a general payoff equivalence result for DU models contained in Volij (2002, 2025).¹¹ In what follows, we present a more direct derivation that is valid as long as g is differentiable and strictly increasing.

Consider a representative buyer i with value $v_i \geq r$. If buyer i were to bid like a type $v'_i \geq r$, while all other buyers followed the equilibrium bidding strategy $b(\cdot)$, buyer i 's ex-post payoff would be:

$$\pi_i(v_{-i}, v_i, v'_i) = \begin{cases} v_i - b(v'_i) & \text{if } v'_i > \max_{j \neq i} v_j, \\ 0 & \text{if } v'_i < \max_{j \neq i} v_j. \end{cases}$$

provided that $b(v'_i) \leq v_i$.¹² It follows from (1) that buyer i 's certainty equivalent from such

¹⁰This probability weighting function is assumed to be common knowledge. It is feasible to allow for heterogenous risk preferences if individual buyers' risk types are common knowledge. Private risk types induce a far more complex game with multidimensional private information. See Che and Gale (2006).

¹¹We thank an anonymous referee for this insight. We would like to point out that this approach of deriving the equilibrium in FPA however relies on the existence of such an equilibrium.

¹²It is clear that v'_i with $b(v'_i) > v_i$ cannot be optimal, since the induced lottery is stochastically dominated by that of a bid with $b(v'_i) \leq v_i$. A buyer is better off losing than winning for a negative gain.

a strategy is:¹³

$$U_g(\pi_i \mid v_i, v'_i) = g([F(v'_i)]^{N-1}) [v_i - b(v'_i)].$$

In equilibrium, truthful revelation must be optimal. Formally, it must be the case that:

$$v_i = \arg \max_{v'_i} g([F(v'_i)]^{N-1}) [v_i - b(v'_i)]. \quad (2)$$

The first-order condition corresponding to (2) is a differential equation:

$$\frac{dg([F(v_i)]^{N-1})}{dv_i} \cdot [v_i - b(v_i)] = g([F(v_i)]^{N-1}) \cdot b'(v_i). \quad (3)$$

Solving this differential equation (3) with boundary condition $b(r) = r$ yields a closed-form equilibrium bidding strategy, as shown in the following proposition.

Proposition 1 *Fix $g \in \mathcal{G}$. In an FPA with reserve price $r \in [\underline{v}, \bar{v}]$, the symmetric increasing equilibrium bidding strategy is given by*

$$b(v_i; r) = v_i - \frac{\int_r^{v_i} g([F(t)]^{N-1}) dt}{g([F(v_i)]^{N-1})}, \quad (4)$$

which increases with r .

If g is star-shaped, weak risk aversion generates “overbidding” in the FPA. In other words, buyers of all types submit a (weakly) higher bid than under risk neutrality. To see this, recall that buyer risk neutrality is represented by the distortion function $\check{g}(x) = x$. Because the probability weighting function $g(\cdot)$ is star-shaped, we have $\frac{g(x)}{g(y)} \leq \frac{x}{y} = \frac{\check{g}(x)}{\check{g}(y)}$, $\forall x < y$. Therefore $b(\cdot; r)$ is greater than the analogous risk neutral bidding strategy:

$$b(v_i; r) \geq v_i - \frac{\int_r^{v_i} [F(t)]^{N-1} dt}{[F(v_i)]^{N-1}}.$$

It is well-known that for risk neutral buyers the seller’s expected revenue is the same in the FPA and SPA for a fixed reserve (the FPA and SPA are “revenue equivalent”). Because buyer risk aversion does not affect equilibrium bidding strategies in the SPA, overbidding in

¹³The conditional cdf of π_i is

$$H(t \mid v_i, v'_i) = \Pr(\pi_i \leq t \mid v_i, v'_i) = \begin{cases} 1 & \text{for } t \geq v_i - b(v'_i), \\ 1 - [F(v'_i)]^{N-1} & \text{for } 0 \leq t < v_i - b(v'_i). \end{cases}$$

the FPA implies that the FPA generates greater expected revenue for the seller. Revenue equivalence is broken in favor of the FPA, as in the EU model.

We can also compare auctions in which buyers exhibit different degrees of risk aversion. To this end, we define the following partial order over $\mathcal{G}$.

Definition 1 For $g, \check{g} \in \mathcal{G}$, we write $\check{g} \geq_* g$ if there exists a star-shaped cdf $h(\cdot)$ defined on $[0, 1]$ and satisfying $h(0) = 0$ and $h(1) = 1$ such that $\check{g}(\cdot) = h(g(\cdot))$. If $h(\cdot)$ is strictly star-shaped (i.e., $h(x)/x$ strictly increases on $[0, 1]$), then we write $\check{g} >_* g$.

If $\check{g} \geq_* g$, then $\check{g}$ can be said to exhibit greater risk aversion. Because $h(\cdot)$ is star-shaped, we have $\frac{\check{g}(x)}{\check{g}(y)} = \frac{h(g(x))}{h(g(y))} \leq \frac{g(x)}{g(y)}$, $\forall x < y$. From Proposition 1, we have the following corollary.

Corollary 1 For $g, \check{g} \in \mathcal{G}^*$ satisfying $\check{g} \geq_* g$,

- (i) Bids in the FPA are greater when buyers are equipped with $\check{g}$ rather than g .
- (ii) The difference in revenue between the FPA and the SPA is greater when buyers are equipped with $\check{g}$ rather than g .

Part (i) implies that greater buyer risk aversion leads to stronger bids. In turn, this implies that the seller's expected revenue is increasing with buyer risk aversion in the FPA (for a fixed r). Part (ii) follows from the observation that SPA revenue is unchanged.

For certain results, it is useful to order preferences by convexity.

Definition 2 For $g, \check{g} \in \mathcal{G}$, we write $\check{g} \geq_{cvx} g$ if there exists a convex cdf $h(\cdot)$ defined on $[0, 1]$ and satisfying $h(0) = 0$ and $h(1) = 1$ such that $\check{g}(\cdot) = h(g(\cdot))$. If $h(\cdot)$ is strictly convex, then we write $\check{g} >_{cvx} g$.

If both g and h are convex, $\check{g}$ is more convex than g . Clearly, $\check{g} \geq_{cvx} g$ implies $\check{g} \geq_* g$.

2.2.1 Incorporating entry fees

We now extend the analysis to the case with weakly positive entry fees $e \geq 0$. For simplicity, we restrict attention to pure-strategy threshold participation strategies. The following lemma characterizes the participation threshold under both formats.

Lemma 1 For an FPA or SPA with reserve price $r \in [\underline{v}, \bar{v}]$ and entry fee $e \in [0, \bar{v} - r]$, potential buyer i participates if and only if $v_i \geq v_{\dagger}(r, e)$, where the equation

$$e = g(F(v_{\dagger}(r, e))^{N-1})(v_{\dagger}(r, e) - r) \quad (5)$$

has a unique solution $v_{\dagger}(r, e) \in [r, \bar{v}]$. For $e > \bar{v} - r$, no buyers participate.

In the SPA, truthful bidding remains a weakly dominant strategy for entrants. In the FPA, the differential equation (3) continues to pin down the bidding strategy among entrants, but the boundary condition is shifted to $b(v_{\dagger}(r, e)) = r$ where $v_{\dagger}(r, e) > r$.

Proposition 2 *Fix $g \in \mathcal{G}$. In an FPA with reserve price $r \in [\underline{v}, \bar{v}]$ and entry fee $e \in [0, \bar{v} - r]$, the symmetric increasing equilibrium bidding strategy is given by*

$$b(v_i; r, e) = v_i - \frac{e + \int_{v_{\dagger}(r, e)}^{v_i} g([F(t)]^{N-1}) dt}{g([F(v_i)]^{N-1})}, \quad (6)$$

which increases with r and decreases with e .

The introduction of a positive entry fee $e > 0$ with a fixed reserve r has two effects. First, a winning buyer's transfer to the seller is partly shifted from the bid to the entry fee. Hence, equilibrium bids fall. Weaker buyers for whom the optimal bid now falls below the reserve are screened out. As a result, inframarginal competition intensifies. These forces are illustrated by the first and second terms on the right hand side of (7), respectively, where

$$b(v_i; r, e) - b(v_i; r, 0) = -\frac{e}{g([F(v_i)]^{N-1})} + \frac{\int_r^{v_{\dagger}(r, e)} g([F(t)]^{N-1}) dt}{g([F(v_i)]^{N-1})} \quad (7)$$

is valid for $v_i \geq v_{\dagger}(r, e)$. The first term is negative; the second is positive.

3 Optimal screening for a risk neutral seller

In this section we consider optimal reserve prices and entry fees for a risk neutral seller faced with risk averse buyers. We first establish that reserve-only screening mechanisms dominate those with positive entry fees, for both the FPA and the SPA. Hence, we can focus on reserve prices. In the remainder of the section, we characterize the optimal reserve prices.

We now state the main dominance result in favor of reserve-only mechanisms.

Proposition 3 *Fix $g_b \in \mathcal{G}^*$. Let $U_s(r, e)$ denote a risk neutral seller's expected payoff in an FPA or SPA with reserve price $r \in [\underline{v}, \bar{v})$ and entry fee $e \in [0, \bar{v} - r]$. Holding fixed the auction format, $U_s(\tilde{r}, 0) \geq U_s(r, e)$ where $\tilde{r} = v_{\dagger}(r, e) \in [r, \bar{v}]$.*

Let v^1 denote the maximum valuation among the N potential buyers and $[F(v^1)]^N$ its distribution. In the FPA, a risk neutral seller's expected payoff under (r, e) is given by

$$U_s^{FPA}(r, e) = v_0 F(\tilde{r})^N + eN \{1 - F(\tilde{r})\} + \int_{\tilde{r}}^{\bar{v}} b(v^1; r, e) d[F(v^1)]^N \quad (8)$$

where $\tilde{r}$ coincides with the participation threshold $v_{\dagger}(r, e)$. This expression consists of three parts. The first term captures the seller's payoff in the state that no potential buyers submit bids above the r . The second is the expected sum of entry fees from participants. The third is the expected payment from the winning buyer, conditional on having $K \geq 1$ participants.

Comparing the equilibrium bidding functions, we see that

$$b(v_i; r, e) = b(v_i; \tilde{r}, 0) - \frac{e}{g_b \left([F(v_i)]^{N-1} \right)} \quad (9)$$

where the notation $b(\cdot; \cdot, \cdot)$ conditions the equilibrium bidding strategy on the screening mechanism. With appropriate substitutions, it follows that:

$$U_s^{FPA}(r, e) = U_s^{FPA}(\tilde{r}, 0) + eN \left[\{1 - F(v_{\dagger})\} - \int_{F(v_{\dagger})}^1 \frac{u^{N-1}}{g_b(u^{N-1})} du \right] \quad (10)$$

Because g_b is star-shaped, $\frac{x}{g(x)} \geq \frac{1}{g(1)} = 1$. Thus, the term in brackets is non-positive. This implies that the reserve-only mechanism is weakly preferred by the seller.

A similar result obtains for the second-price auction. In this case,

$$U_s^{SPA}(r, e) = U_s^{SPA}(\tilde{r}, 0) + eN \left(1 - \frac{F(\tilde{r})^{N-1}}{g_b \left(F(\tilde{r})^{N-1} \right)} \right) \cdot \{1 - F(\tilde{r})\} \quad (11)$$

Star-shapedness of g_b implies that the term in parentheses is non-positive.

Proposition 3, together with our reserve price characterizations below, implies that the reserve-only FPA is optimal among standard FPA and SPA auctions in which the seller is restricted to reserve prices and entry fees. In fact, this conclusion follows directly from Gershkov et al. (2020), who establish the optimality of the reserve-only FPA among no-insurance mechanisms for a risk neutral seller when DU buyers are risk averse.¹⁴

Though we focus on the case that g is weakly or strongly risk averse, our equilibrium characterization is valid for any $g \in \mathcal{G}$. If $g(x) \geq x$ for all $x \in [0, 1]$ (e.g., if g is concave) then the above arguments imply that fee-only mechanisms are weakly preferred by the seller. Such distortions represent risk-seeking behavior.

3.1 Optimal reserve prices

We now restrict attention to mechanisms without entry fees.

Because risk aversion does not affect buyers' strategies in the SPA, the seller's optimal reserve is the same as if buyers were risk neutral. Recall that v_0 is the seller's value. In

¹⁴No-insurance mechanisms are those in which losing bidders do not receive transfers from the seller.

the case that buyers are risk neutral, it is well-known that the optimal reserve price is $r^M := J^{-1}(v_0)$ under our assumption that $J(r) = r - \frac{1-F(r)}{f(r)}$ is strictly increasing (Myerson, 1981). When buyers are risk averse, the optimal reserve price is again r^M in the SPA.

In the FPA, increasing the reserve induces more aggressive bidding among participants but increases the probability of a failed sale. The latter effect (a marginal cost) is independent of buyers' risk preferences. The former is not: from Proposition 1 and the star-shapedness of g_b , we have $\frac{\partial b(v_i; r)}{\partial r} = \frac{g_b([F(r)]^{N-1})}{g_b([F(v_i)]^{N-1})} \leq \frac{[F(r)]^{N-1}}{[F(v_i)]^{N-1}}$. Hence, a higher reserve does less to stimulate aggressive bidding when buyers are weakly risk averse for the same marginal cost. As we now show, the optimal reserve is therefore lower when buyers are weakly risk averse.

Interpreting v_0 as a monetary payoff, let $R^{FPA}(r) := U_s^{FPA}(r, 0)$ denote expected revenue and consider the seller's optimal choice of r . Differentiating with respect to r , we obtain

$$\frac{dR^{FPA}(r)}{dr} = (v_0 - r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} \frac{g_b([F(r)]^{N-1})}{g_b([F(v^1)]^{N-1})} d[F(v^1)]^N \quad (12)$$

If an interior solution exists, it satisfies $\frac{dR^{FPA}(r)}{dr} = 0$. This equation has a unique root only if $\frac{dR^{FPA}(r)}{dr}$ single-crosses zero. For example, when $g_b(x) = x$ is the identity function such that buyers are risk neutral, then

$$\frac{dR^{FPA}(r)}{dr} \Big|_{g(x)=x} = N [F(r)]^{N-1} f(r) \left\{ v_0 - \left(r - \frac{1 - F(r)}{f(r)} \right) \right\} \quad (13)$$

which crosses zero once from above under the maintained assumption that $J(\cdot)$ is strictly increasing. The solution in this case is the Myerson reserve r^M .

The following lemma shows that buyer risk aversion reduces the marginal benefit of increasing the reserve price relative to risk neutral case.

Lemma 2 *In general,*

$$\frac{dR^{FPA}(r)}{dr} \leq \frac{dR^{FPA}(r)}{dr} \Big|_{g_b(x)=x}. \quad (14)$$

where the inequality is strict if g_b is strictly star-shaped.

Lemma 2 implies that when bidders are strictly risk averse, the seller's optimal reserve price in the FPA is strictly below the Myerson reserve, and hence lower than the optimal reserve price in the SPA. The next result generalizes this insight to show that under certain restrictions, the FPA optimal reserve is lower whenever buyers are more risk averse.

Proposition 4 *Fix $\check{g}_b, g_b \in \mathcal{G}^*$ with $\check{g}_b \geq_* g_b$. If $\frac{dR^{FPA}(r)}{dr} \Big|_{\check{g}_b}$ and $\frac{dR^{FPA}(r)}{dr} \Big|_{g_b}$ each cross zero once, $r^{FPA} \Big|_{\check{g}_b} \leq r^{FPA} \Big|_{g_b}$. If $\check{g}_b >_* g_b$, the inequality is strict.*

For an arbitrary $g_b \in \mathcal{G}^*$, $J(x)$ strictly increasing is no longer sufficient for single crossing. For example, with the power distortion $g_b(x) = x^\gamma$ for $\gamma > 0$ and $\gamma \neq N/(N-1)$, the analogous requirement is $J_\gamma(x)$ strictly increasing, where:

$$J_\gamma(x) = x - \left\{ \frac{1}{N - \gamma \cdot (N-1)} \right\} \left[\frac{F(x)^{(\gamma-1)(N-1)} - F(x)}{f(x)} \right] \quad (15)$$

If $F(\cdot)$ is known, this new requirement can be checked directly. For example, if $F(\cdot)$ is uniform on $[0, 1]$ and $v_0 = 0$, $J_\gamma(x) - v_0$ single-crosses zero when $\gamma < (N+1)/(N-1)$. In empirical applications, it is possible to estimate $F(\cdot)$ nonparametrically.¹⁵

In the risk neutral setting ($\gamma = 1$), it is common to impose log-concavity of the valuation density $f(\cdot)$. Log-concavity implies $J'_1(x) \geq 1$, and hence single-crossing, and is satisfied by many common parametric distributions. When $\gamma \neq 1$, however, log-concavity is no longer sufficient in general for single-crossing – even for small deviations from risk neutrality.¹⁶

Provided that a unique optimum r^{FPA} exists, Gershkov et al.(2020) show that it satisfies:

$$r = v_0 + \frac{1}{f(r)} \left[\int_{F(r)}^1 \frac{t^{N-1}}{F(r)^{N-1}} \frac{g_b(F(r)^{N-1})}{g_b(t^{N-1})} dt \right] \quad (16)$$

This formulation makes clear the connection to the Myerson optimal reserve. When g_b is the identity function the term in brackets simplifies to $1 - F(r)$. For $g_b \in \mathcal{G}^*$, $\frac{y g(x)}{x g(y)} \leq 1$ for $y \geq x$, so the term in brackets is weakly smaller than $1 - F(r)$. The right hand side is therefore rotated downwards compared with the risk neutral case, implying $r^{FPA} \leq r^M$.

For applied work in which it is desirable to compute the optimal reserve (e.g., in a counterfactual), it is significant that (16) takes a closed form. In the EU model, no such closed form is available; in order to compute the optimal reserve, one must generally solve a series of (often delicate) differential equations, one for each candidate r .

4 Optimal screening for a risk averse seller

We now consider the optimal choice of entry fees and reserve prices for a risk averse seller. We first prove a general result which implies that reserve-only screening mechanisms are weakly dominant for strongly risk averse sellers. We then specialize this result to the case of DU risk averse sellers with convex distortion $g_s \in \mathcal{G}^{cvx}$ in order to characterize the optimal reserve prices in the FPA and SPA. Finally, we compare the two. As before, we assume that all buyers are equipped with a symmetric probability weighting function $g_b \in \mathcal{G}^*$.

¹⁵In Li et al. (2026), we establish nonparametric identification of $F(\cdot)$.

¹⁶In this case we further require that that $|\partial_\gamma J'_\gamma(x)| \leq L < \infty$ in a neighborhood of $\gamma = 1$ near r^M .

Proposition 5 Fix $g_b \in \mathcal{G}^*$. Let $\Pi(r, e)$ denote the seller's payoff distribution in an FPA or SPA with reserve price $r \in [\underline{v}, \bar{v})$ and entry fee $e \in [0, \bar{v} - r]$. Holding fixed the auction format, $\Pi(r, e)$ is second order stochastically dominated by $\Pi(\tilde{r}, 0)$ where $\tilde{r} = v_+(r, e)$.

Corollary 2 If the seller is a DU maximizer with $g_s \in \mathcal{G}^{cvx}$, then $U_s(\tilde{r}, 0) \geq U_s(r, e)$, where $U_s(r, e)$ denotes the seller's certainty equivalent given mechanism (r, e) .

For the SPA, the proof of Proposition 5 is particularly straightforward. Let v^1 and v^2 denote the first and second highest valuations, respectively, and let k the realized number of bidders. Given (v^1, v^2) , the difference in the seller's payoffs between $(\tilde{r}, 0)$ and (r, e) is:

$$\pi(k; \tilde{r}, 0) - \pi(k; r, e) = \begin{cases} 0 & \text{if } k = 0 \\ \left(\frac{1}{g_b(F(\tilde{r})^{N-1})} - 1 \right) e & \text{if } k = 1 \\ -ke & \text{if } k \geq 2 \end{cases} \quad (17)$$

Intuitively, the entry fee transfers a component of the payoff from the $k = 1$ state to the $k \geq 2$ states: a reserve-only mechanism increases the seller's payoff when $k = 1$, but reduces it when $k \geq 2$. As a result, the cdf of the payoff distribution under $(\tilde{r}, 0)$ crosses the cdf of the payoff distribution under (r, e) once from below. From Proposition 3, the former has a greater expectation. These facts together imply second order stochastic dominance.

For the first-price auction, we make a two part argument. We first observe that the distribution of payoffs induced by (r, e) is a mean preserving spread the distribution of expected payoffs conditional on v^1 . The latter is second order stochastically dominated by the distribution of payoffs induced by $(\tilde{r}, 0)$. This can be shown by comparing the integrated quantile functions associated with each distribution conditional on $K \geq 1$.

By Proposition 5, any seller with preferences that respect second order stochastic dominance prefers the reserve-only mechanism.¹⁷ Convexity of g_s implies that the seller's preferences respect second order stochastic dominance. Hence, Corollary 2 follows immediately.

Waehrer et al. (1998) establish a similar result for the FPA with an EU risk averse seller and risk neutral buyers (see Theorem 4). To our knowledge, there are no analogous results in the literature when buyers and sellers are both potentially risk averse. Examining optimal reserve prices in such an environment (under expected utility), Hu et al. (2010) speculate that the effect of entry fees could be ambiguous: "if the seller is able to charge bidders an entry fee, he may wish to do so if the bidders are risk averse [Maskin and Riley (1984)], but not if he is risk averse and can also set the reserve price [Waehrer et al. (1998)]. The nature of optimal combinations of entry fees and reserves when the seller or buyers are risk averse is unknown." Our results hence provide one possible resolution to this question.

¹⁷Such as EU preferences with concave von Neumann-Morgenstern utility or various NEU preferences.

4.1 Optimal reserve prices

It remains to characterize the optimal reserve prices with a risk averse seller.

4.1.1 Second-price auction

We first prove that the reserve prices below v_0 are never optimal for the seller. To this end, we show that the payoff distribution induced by a reserve price $r < v_0$ is stochastically dominated by the payoff distribution induced by $r = v_0$. For any $g_s \in \mathcal{G}$, this implies that we can restrict our search for the optimal reserve to $[v_0, \bar{v}]$.

Lemma 3 *Fix $g_s \in \mathcal{G}$. For any $r \in [v, v_0)$, $U^{SPA}(r, 0) < U^{SPA}(v_0, 0)$.*

Denote the distribution of the second-highest buyer value by $\tilde{F}(v) = N[F(v)]^{N-1}(1 - F(v)) + [F(v)]^N$. For $r > v_0$, the distribution of seller payoffs is given by

$$H_s(t|r) = \Pr(\pi_0 \leq t|r) = \begin{cases} \tilde{F}(t) & \text{for } t \in [r, \bar{v}], \\ [F(r)]^N & \text{for } t \in [v_0, r), \\ 0 & \text{for } t \in [0, v_0). \end{cases}$$

As buyers bid truthfully in the SPA, the payoff distribution does not depend on buyers' distortion g_b . Hence, buyer risk aversion does not affect the optimal reserve in the SPA.

We next derive the marginal change in the seller's certainty equivalent $U^{SPA}(r) := U^{SPA}(r, 0)$ from increasing r . Under the assumption of strict risk aversion, this marginal benefit term is strictly negative above the Myerson optimal reserve r^M provided that $J(\cdot)$ is strictly increasing. This implies that the optimal reserve is contained in $[v_0, r^M)$.

Lemma 4 *Fix $g_s \in \mathcal{G}^{cvx}$. For all $r \in (v_0, \bar{v})$,*

$$\partial_r U^{SPA}(r) = g'_s(1 - [F(r)]^N)N[F(r)]^{N-1}f(r)(v_0 - r) \quad (18)$$

$$\begin{aligned} & + g_s(1 - [F(r)]^N) - g_s(1 - \tilde{F}(r)) \\ & \leq g'_s(1 - [F(r)]^N)N[F(r)]^{N-1}f(r)[v_0 - J(r)]. \end{aligned} \quad (19)$$

and (19) is zero when $r = r^M$. If g_s is strictly convex, the inequality holds strictly, and the seller-optimal reserve price in the SPA is strictly lower than r^M .

Lemma 4 provides a partial DU counterpart to Hu et al. (2010)'s Theorem 2 for the EU environment, which shows that the SPA and FPA optimal reserves are both decreasing in the seller's risk aversion.¹⁸ In the SPA, increasing the reserve price raises the expected

¹⁸We address the FPA case in the next subsection.

payment if a sale occurs but decreases the probability of a sale. EU risk averse sellers derive lower utility from high payments when a sale occurs, while DU risk averse sellers overweight the probability that a sale will not occur. In each case, a lower reserve is optimal.

4.1.2 First-price auction

We first argue that the optimal reserve price must be weakly greater than the seller's value v_0 . Compared to $r = v_0$, a lower reserve decreases the bid for every type by Proposition 1 while increasing the chance of a sale below v_0 . Hence, the seller always prefers $r \geq v_0$.

We next examine the marginal benefit of increasing the reserve in the case that $r \geq v_0$. The distribution of the seller's payoffs conditional on r is now given by

$$H_s(t|r) = \Pr(\pi_0 \leq t|r) = \begin{cases} [F(b^{-1}(t; r))]^N & \text{for } t \in [r, b(\bar{v}; r)], \\ [F(r)]^N & \text{for } t \in [v_0, r), \\ 0 & \text{for } t \in [0, v_0). \end{cases}$$

One difference between the FPA and the SPA is that the payoff distribution, and hence the optimal reserve, now generally depends on buyers' risk preferences through the equilibrium bidding function. The following lemma provides a parallel result to Lemma 4 above: the marginal benefit of increasing the reserve is strictly negative at r^M if g_s is strictly convex.

Lemma 5 *Fix $g_b \in \mathcal{G}^*$ and $g_s \in \mathcal{G}^{cvx}$. For all $r \in [v_0, \bar{v})$,*

$$\begin{aligned} \partial_r U^{FPA}(r) &= g'_s(1 - [F(r)]^N)N[F(r)]^{N-1}f(r)(v_0 - r) \\ &\quad + \int_r^{\bar{v}} \frac{g_b([F(r)]^{N-1})}{g_b([F(v)]^{N-1})} g'_s(1 - [F(v)]^N)N[F(v)]^{N-1}f(v)dv \end{aligned} \quad (20)$$

$$\leq g'_s(1 - [F(r)]^N)N[F(r)]^{N-1}f(r)[v_0 - J(r)]. \quad (21)$$

and (21) is zero when $r = r^M$. If g_s is strictly convex, the inequality holds strictly, and the seller-optimal reserve price in the FPA is strictly lower than r^M .

Like in the SPA, the marginal benefit of increasing the reserve is strictly negative above the Myerson optimal reserve r^M for strictly convex g_s provided that $J(\cdot)$ is strictly increasing. This implies that the optimal reserve is contained in $[v_0, r^M)$.

It is interesting to analyze how the optimal reserve price varies with buyer and seller risk aversion. As in Section 3.1, $J(\cdot)$ strictly increasing does not guarantee that $\partial_r U^{FPA}(r)$ crosses zero once on $[v_0, r^M)$. Hence, (20) may not have a unique root. However, if $g_s, \check{g}_s \in \mathcal{G}^{cvx}$ with $g_s \geq_{cvx} \check{g}_s$ and if $\partial_r U^{FPA}(r)|_{g_s}$ and $\partial_r U^{FPA}(r)|_{\check{g}_s}$ both cross zero once from above on $[v_0, r^M)$, then it follows from (20) that a seller equipped with g_s prefers a lower reserve than a seller

equipped with $\check{g}_s$. Similarly, if $g_b, \check{g}_b \in \mathcal{G}^*$ with $g_b \geq_* \check{g}_b$ and if $\partial_r U^{FPA}(r)|_{g_b}$ and $\partial_r U^{FPA}(r)|_{\check{g}_b}$ both cross zero once from above on $[v_0, r^M]$, a seller prefers a lower reserve when bidders are equipped with g_b rather than $\check{g}_b$. Put differently: greater seller risk aversion (in the strong sense) and greater buyer risk aversion (in the weak sense) each result in lower reserves.

4.2 Comparison of first- and second-price auctions

Our first main result in this section shows that the seller-optimal reserve is lower in the FPA than in the SPA under a mild condition on the seller's marginal benefit functions and a restriction on g_s . This result is analogous to Theorem 1 of Hu et al. (2010).

Proposition 6 *If $g_s \in \mathcal{G}^{cvx}$ is strictly convex and both $\partial_r U^{FPA}(r)$ and $\partial_r U^{SPA}(r)$ cross zero once, then the optimal reserve in the FPA is strictly lower than in the SPA.*

To prove this result, it suffices to show that the seller's marginal benefit from increasing the reserve price is strictly smaller in the FPA than in the SPA, $\partial_r U^{FPA}(r) \leq \partial_r U^{SPA}(r)$. Comparing Lemmas 4 and 5, we see that this is the case whenever

$$N [F(r)]^{N-1} \int_r^{\bar{v}} g'_s(1 - [F(v)]^N) dF(v) \leq g_s(1 - [F(r)]^N) - g_s(1 - \tilde{F}(r)). \quad (22)$$

In the FPA, a greater reserve increases all inframarginal bids, including the winning bid, via Proposition 1. This benefit is represented by the left hand side of (22). In the SPA, a greater reserve increases the winner's payment in the event that exactly one potential buyer submits a bid above the reserve. This is represented on the right hand side of (22). The next lemma shows that the inequality is satisfied whenever the seller's distortion is convex.

Lemma 6 *Inequality (22) holds weakly for any $g_b \in \mathcal{G}^*$ and $g_s \in \mathcal{G}^{cvx}$ and strictly if g_s is also strictly convex. Thus, $\partial_r U^{FPA}(r) < \partial_r U^{SPA}(r)$ if g_s is strictly convex.*

We prove this result by demonstrating that (22) is implied by an integral inequality

$$\int_a^b \{g'_s(s) - g'_s(a)\} \cdot \left(1 - \frac{F(r)^{N-1}}{(1-s)^{(N-1)/N}}\right) ds \geq 0 \quad (23)$$

for constants $a \leq b$ that depend on $F(r)$. As shown in the proof, the term in parentheses is non-negative, so that inequality (23) is satisfied for any convex g_s . Inequality (23) holds strictly when g_s is strictly convex because $(1-s)^{(N-1)/N} > F(r)^{N-1}$ for all $s \in [a, b]$.

The prior result establishes that DU risk averse sellers prefer different reserve prices depending on the auction format. We now compare the two formats at a common reserve. We first prove a general stochastic ordering result before specializing to DU sellers.

Proposition 7 *Fix $g_b \in \mathcal{G}^*$. For a fixed reserve $r \in [v_0, \bar{v})$, the distribution of SPA payoffs is second order stochastically dominated by the distribution of FPA payoffs.*

Corollary 3 *If the seller is a DU maximizer with $g_s \in \mathcal{G}^{cvx}$, then $U_s^{FPA}(r) \geq U_s^{SPA}(r)$. If g_b is strictly star-shaped or g_s is strictly convex, the inequality is strict.*

Corollary 3 is analogous to Hu et al. (2010) Proposition 4(i) for the EU environment. Both results share a common underlying mechanism. For weakly risk averse DU buyers, this mechanism is captured in Proposition 7. Consider an auction with a fixed reserve r . With risk neutral buyers, the FPA and SPA generate the same payoff in expectation (“revenue equivalence”). However, SPA payoffs are a mean preserving spread of FPA payoffs. A strongly risk averse seller therefore prefers the FPA – regardless of whether strong risk aversion arises from EU, DU, or other risk preferences. For $g_b \in \mathcal{G}^*$, revenue equivalence is broken in favor of the FPA, reinforcing the seller’s preference for the FPA.

An immediate implication of Proposition 7 is that strongly risk averse sellers prefer the optimal FPA to the optimal SPA. Take the case of a DU risk averse seller with $g_s \in \mathcal{G}^{cvx}$. If r^{FPA} is the FPA optimal reserve and r^{SPA} is the SPA optimal reserve, then

$$U_s^{FPA}(r^{FPA}) \geq U_s^{FPA}(r^{SPA}) \geq U_s^{SPA}(r^{SPA}). \quad (24)$$

where the second inequality holding strictly given strict buyer or seller risk preferences. A strongly risk averse DU seller prefers the FPA whenever buyers are weakly risk averse.

5 Conclusion

In this paper, we study seller-optimal screening mechanisms in first- and second-price auctions when agents have preferences described by Yaari (1987)’s dual utility model. Our main contribution is to show that risk averse sellers prefer reserve-only mechanisms when buyers are weakly risk averse DU maximizers. In particular, the result applies for risk neutral buyers. The key requirement for our result is that the seller’s preferences respect second order stochastic dominance. This property is satisfied by many types of EU and non-EU preferences besides DU. As such, our result implies more generally that reserve-only mechanisms are preferred by strongly risk averse sellers when buyers are risk neutral. In this sense, our result generalizes the EU dominance result contained in Waehrer et al (1998).

Our second key result shows that strongly risk averse sellers prefer the optimal reserve-only FPA to the optimal reserve-only SPA when buyers are weakly risk averse DU maximizers. We characterize the optimal reserve prices directly in the case that the seller is a strongly risk averse DU maximizer. We show that optimal reserve prices are lower in the FPA than in

the SPA and decreasing in seller risk aversion under certain regularity conditions. Moreover, the FPA optimal reserve decreases in buyer risk aversion. These results are broadly similar to those of Hu et al. (2010), despite the differences between the DU and EU models. Our work and that of Hu et al. (2010) might be integrated by adopting the rank-dependent expected utility (RDEU) model of Quiggin (1982), which combines nonlinear probability-weighting with nonlinear payoff-weighting. We leave this investigation to future work.

With DU risk averse buyers, Gershkov et al. (2022) show that the revenue-maximizing auction entails participation subsidies. The optimal FPA in the sense considered in our paper is therefore suboptimal for a risk neutral seller. We thus anticipate that our optimal FPA is suboptimal among all possible mechanisms available to a risk averse DU seller. Nevertheless, our results have significant practical relevance: sellers in most real-world auction environments are limited to standard auction formats and simple screening mechanisms, with the FPA and SPA being especially common. Explicit participation subsidies are rare.

Two strong assumptions in our paper are that buyers' valuations are independent and that buyers share a common distortion function. When these assumptions are plausible, our results recommend the reserve-only FPA to maximize the seller's certainty equivalent. One way to extend the analysis is by allowing for affiliation among potential buyers' private valuations, as in Waehrer et al. (1998). Relaxing the common distortion function assumption raises more serious difficulties, akin to those discussed in Che and Gale (2006): if buyers' risk preferences are private information, even simple auction formats such as the FPA and SPA engender complex games of multidimensional private information.

A Appendix

The Appendix is organized into three parts. The first part contains results related to the derivation of the bidding equilibrium in Section 2. The second part contains all results related to the characterization of optimal reserve prices. The third part contains the entry fee dominance results, which are the more technically involved.

A.1 Bidding Equilibrium Results

Proof of Propositions 1 and 2

We present one proof as Proposition 1 is implied by Proposition 2 for $e = 0$. From (3),

$$\begin{aligned} \frac{d \left\{ g([F(v_i)]^{N-1})b(v_i) \right\}}{dv_i} &= \frac{dg([F(v_i)]^{N-1})}{dv_i} b(v_i) + g([F(v_i)]^{N-1}) \cdot b'(v_i) \\ &= \frac{dg([F(v_i)]^{N-1})}{dv_i} b(v_i) + \frac{dg([F(v_i)]^{N-1})}{dv_i} \cdot [v_i - b(v_i)] = \frac{dg([F(v_i)]^{N-1})}{dv_i} \cdot v_i. \end{aligned} \quad (25)$$

Only buyers with values above $v_{\dagger}$ submit bids. In equilibrium, $b(v_{\dagger}) = r$. Integrating both sides of (25) over $[v_{\dagger}, v_i]$, we have

$$\int_{v_{\dagger}}^{v_i} \frac{d \left\{ g([F(t)]^{N-1})b(t) \right\}}{dt} dv_i = \int_{v_{\dagger}}^{v_i} \frac{dg([F(t)]^{N-1})}{dt} \cdot t dt,$$

which gives

$$\begin{aligned} \int_{v_{\dagger}}^{v_i} d \left\{ g([F(t)]^{N-1})b(t) \right\} &= \int_{v_{\dagger}}^{v_i} t dg([F(t)]^{N-1}), \\ \Leftrightarrow g([F(t)]^{N-1})b(t)|_{v_{\dagger}}^{v_i} &= t g([F(t)]^{N-1})|_{v_{\dagger}}^{v_i} - \int_{v_{\dagger}}^{v_i} g([F(t)]^{N-1}) dt, \\ \Leftrightarrow g([F(v_i)]^{N-1})b(v_i) - g([F(v_{\dagger})]^{N-1})r &= v_i g([F(v_i)]^{N-1}) - v_{\dagger} g([F(v_{\dagger})]^{N-1}) \\ &\quad - \int_{v_{\dagger}}^{v_i} g([F(t)]^{N-1}) dt, \\ \Leftrightarrow g([F(v_i)]^{N-1})b(v_i) &= v_i g([F(v_i)]^{N-1}) - (v_{\dagger} - r) g([F(v_{\dagger})]^{N-1}) \\ &\quad - \int_{v_{\dagger}}^{v_i} g([F(t)]^{N-1}) dt, \\ \Leftrightarrow b(v_i; r, e) &= v_i - \frac{e + \int_{v_{\dagger}}^{v_i} g([F(t)]^{N-1}) dt}{g([F(v_i)]^{N-1})}, \forall v_i \geq v_{\dagger}. \end{aligned}$$

This gives the closed-form equilibrium bidding strategy. For any strictly increasing g ,

$\frac{\partial^2 U_g(\pi_i|v_i, v'_i)}{\partial v_i \partial v'_i} = g'([F(v'_i)]^{N-1})(N-1)[F(v'_i)]^{N-2}f(v'_i) > 0$. This guarantees global incentive compatibility. For the zero fee case $e = 0$, note that $v_{\dagger} = r$, which implies the bidding strategy from Proposition 1.

Proof of Lemma 1

It is clear a potential buyer with valuation v_i never gains from participating if $v_i - r < e$. Thus, if $e > \bar{v} - r$, no potential buyers gain from participating. This proves the last statement.

Now consider entry fees $e \in [0, \bar{v} - r]$. Suppose $v_{\dagger}$ is the lowest type that enters given (r, e) . Because the distribution of types is continuous, ties occur with probability zero. Hence, a participating buyer with type $v_{\dagger}$ wins only in the event that the other $N - 1$ potential buyers stay out. In the SPA, the payment in this case is the reserve price r . In the FPA, the lowest entering type bids r and hence pays r . Thus, for both formats, participation by the threshold type is associated with the binary lottery

$$\pi_i(v_{\dagger}) = \begin{cases} -e + v_{\dagger} - r & \text{w.p. } F(v_{\dagger})^{N-1} \\ -e & \text{w.p. } 1 - F(v_{\dagger})^{N-1}. \end{cases}$$

By (1), the certainty equivalent of this lottery is

$$-e + g\left(F(v_{\dagger})^{N-1}\right)(v_{\dagger} - r). \quad (26)$$

It follows immediately that $v_{\dagger}$ is the threshold type iff (5) holds.

It remains to show that the solution to (5) is unique. By construction of g and F , the function $x \rightarrow g\left(F(x)^{N-1}\right)(x - r)$ is continuous and strictly increasing on $[r, \bar{v}]$ for every $r \geq \underline{v}$. Moreover, $g\left(F(r)^{N-1}\right)(r - r) = 0$ and $g\left(F(\bar{v})^{N-1}\right)(\bar{v} - r) = \bar{v} - r$ because $g(1) = 1$. Hence, for every $e \in [0, \bar{v} - r]$, there exists a unique type $v_{\dagger} \in [r, \bar{v}]$ such that (26) is zero.

A.2 Optimal Reserve Characterization

Proof of Lemma 2

Note that

$$\begin{aligned}
\frac{dR^{FPA}(r)}{dr} &= v_0 N [F(r)]^{N-1} f(r) - b(r; r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} \frac{\partial b(v^1; r)}{\partial r} d [F(v^1)]^N \\
&= (v_0 - r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} \frac{\partial b(v^1; r)}{\partial r} d [F(v^1)]^N \\
&= (v_0 - r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} \frac{g_b([F(r)]^{N-1})}{g_b([F(v^1)]^{N-1})} d [F(v^1)]^N. \tag{27}
\end{aligned}$$

When $g_b(x) = x$, we have

$$\begin{aligned}
\frac{dR^{FPA}(r)}{dr} \Big|_{g(x)=x} &= (v_0 - r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} N [F(r)]^{N-1} dF(v^1) \\
&= (v_0 - r) N [F(r)]^{N-1} f(r) + N [F(r)]^{N-1} [1 - F(r)] \\
&= N [F(r)]^{N-1} f(r) \left\{ v_0 - \left(r - \frac{1 - F(r)}{f(r)} \right) \right\}.
\end{aligned}$$

For star-shaped g_b , we have $\frac{g_b(x)}{g_b(y)} \leq \frac{x}{y}, \forall x < y$. From (27), we have

$$\begin{aligned}
\frac{dR^{FPA}(r)}{dr} &= (v_0 - r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} \frac{g_b([F(r)]^{N-1})}{g_b([F(v^1)]^{N-1})} d [F(v^1)]^N \\
&\leq (v_0 - r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} \frac{[F(r)]^{N-1}}{[F(v^1)]^{N-1}} d [F(v^1)]^N \\
&= (v_0 - r) N [F(r)]^{N-1} f(r) + \int_r^{\bar{v}} N [F(r)]^{N-1} dF(v^1) \\
&= N [F(r)]^{N-1} f(r) \left\{ v_0 - \left(r - \frac{1 - F(r)}{f(r)} \right) \right\} = \frac{dR^{FPA}(r)}{dr} \Big|_{g_b(x)=x}.
\end{aligned}$$

For strictly star-shaped g_b , $\frac{g_b(x)}{g_b(y)} < \frac{x}{y}, \forall x < y$ and so the inequality is strict.

Proof of Proposition 4

Fix r and observe that

$$\frac{dR^{FPA}(r)}{dr} \Big|_{\check{g}_b} - \frac{dR^{FPA}(r)}{dr} \Big|_{g_b} = \int_r^{\bar{v}} \left\{ \frac{\check{g}_b([F(r)]^{N-1})}{\check{g}_b([F(v^1)]^{N-1})} - \frac{g_b([F(r)]^{N-1})}{g_b([F(v^1)]^{N-1})} \right\} d [F(v^1)]^N \tag{28}$$

As noted in the text, $\check{g}_b \geq_* g_b$ implies $\frac{\check{g}_b(x)}{\check{g}_b(y)} \leq \frac{g_b(x)}{g_b(y)}$ for $y > x$. The term in braces is therefore non-positive. For $\check{g}_b >_* g_b$ the inequality is strict.

If $\left. \frac{dR^{FPA}(r)}{dr} \right|_{\check{g}_b}$ and $\left. \frac{dR^{FPA}(r)}{dr} \right|_{g_b}$ each cross zero once, then the former must be non-positive at $r^{FPA}|_{g_b}$. Hence, $r^{FPA}|_{\check{g}_b} \leq r^{FPA}|_{g_b}$. In the case that $\check{g}_b >_* g_b$ the comparison is strict.

Proof of Lemma 3

Let v_1 and v_2 denote the highest and second-highest realized valuations. For any $r \in [\underline{v}, \bar{v}]$,

$$\pi_r(v_1, v_2) = \begin{cases} v_0 & \text{if } v_1 < r \\ r & \text{if } v_2 < r \wedge v_1 \geq r \\ v_2 & \text{if } v_2 \geq r \end{cases} \quad (29)$$

Fix $\tilde{r} \in [\underline{v}, v_0]$. Then

$$\pi_{v_0}(v_1, v_2) - \pi_{\tilde{r}}(v_1, v_2) = \begin{cases} v_0 - \tilde{r} & \text{if } v_1 \geq v_0 \wedge v_2 \leq \tilde{r} \\ v_0 - v_2 & \text{if } v_1 \geq v_0 \wedge v_2 \in (\tilde{r}, v_0] \\ 0 & \text{otherwise} \end{cases}$$

which is non-negative for every realization of (v_1, v_2) . Hence, the payoff distribution induced by setting $r = v_0$ first order stochastically dominates the payoff distribution induced by setting $r = \tilde{r}$. Dual utility preferences respect monotonicity (Yaari, 1987 Axiom A4).

Proof of Lemma 4

The seller's certainty equivalent is given by

$$U_s^{SPA}(r) = \int_0^{v_0} g_s(1)dt + \int_{v_0}^r g_s(1 - [F(r)]^N)dt + \int_r^{\bar{v}} g_s(1 - \tilde{F}(t))dt \quad (30)$$

Taking the derivative with respect to r ,

$$\begin{aligned}
\partial_r U_s^{SPA}(r) &= \frac{d \left[\int_0^{v_0} g_s(1) dt + \int_{v_0}^r g_s(1 - [F(r)]^N) dt + \int_r^{\bar{v}} g_s(1 - \tilde{F}(t)) dt \right]}{dr} \\
&= g_s(1 - [F(r)]^N) - g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r)(r - v_0) - g_s(1 - \tilde{F}(r)) \\
&= g_s(1 - [F(r)]^N) - g_s(1 - \tilde{F}(r)) - g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r)(r - v_0) \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r)(v_0 - r) + g_s(1 - [F(r)]^N) - g_s(1 - \tilde{F}(r)) \\
&\leq g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r)(v_0 - r) + g'_s(1 - [F(r)]^N) \left(\tilde{F}(r) - [F(r)]^N \right) \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r)(v_0 - r) + g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} (1 - F(r)) \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) \left[v_0 - \left(r - \frac{1 - F(r)}{f(r)} \right) \right]. \tag{31}
\end{aligned}$$

For strictly convex g_s , the inequality holds strictly.

Proof of Lemma 5

When $r \geq v_0$, the distribution of the seller's payoff π_0 is

$$H_s(t|r) = \Pr(\pi_0 \leq t|r) = \begin{cases} [F(b^{-1}(t; r))]^N & \text{for } t \in [r, b(\bar{v}; r)], \\ [F(r)]^N & \text{for } t \in [v_0, r), \\ 0 & \text{for } t \in [0, v_0). \end{cases}$$

We now derive the seller's certainty equivalent:

$$\begin{aligned}
U_s^{FPA}(r) &= \int_0^{v_0} g_s(1) dt + \int_{v_0}^r g_s(1 - [F(r)]^N) dt + \int_r^{b(\bar{v})} g_s(1 - [F(b^{-1}(t))])^N dt \\
&= v_0 + g_s(1 - [F(r)]^N)(r - v_0) + \int_r^{\bar{v}} g_s(1 - [F(v)]^N) b'(v) dv \\
&= v_0 + g_s(1 - [F(r)]^N)(r - v_0) + \int_r^{\bar{v}} g_s(1 - [F(v)]^N) db(v) \\
&= v_0 + g_s(1 - [F(r)]^N)(r - v_0) + g_s(1 - [F(v)]^N) b(v) \Big|_r^{\bar{v}} - \int_r^{\bar{v}} b(v) dg_s(1 - [F(v)]^N) \\
&= v_0 + g_s(1 - [F(r)]^N)(r - v_0) - g_s(1 - [F(r)]^N) b(r) - \int_r^{\bar{v}} b(v) dg_s(1 - [F(v)]^N) \\
&= v_0 + g_s(1 - [F(r)]^N)(r - v_0) - g_s(1 - [F(r)]^N) r - \int_r^{\bar{v}} b(v) dg_s(1 - [F(v)]^N) \\
&= v_0 - g_s(1 - [F(r)]^N) v_0 - \int_r^{\bar{v}} b(v) dg_s(1 - [F(v)]^N). \tag{32}
\end{aligned}$$

Taking the derivative with respect to r :

$$\begin{aligned}
\partial_r U_s^{FPA}(r) &= \frac{d \left[v_0 - g_s(1 - [F(r)]^N) v_0 - \int_r^{\bar{v}} b(v; r) dg_s(1 - [F(v)]^N) \right]}{dr} \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) v_0 - b(r) g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) \\
&\quad - \int_r^{\bar{v}} \frac{\partial b(v; r)}{\partial r} dg_s(1 - [F(v)]^N) \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) v_0 - r g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) \\
&\quad - \int_r^{\bar{v}} \frac{g([F(r)]^{N-1})}{g([F(v)]^{N-1})} dg_s(1 - [F(v)]^N) \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) (v_0 - r) \\
&\quad + \int_r^{\bar{v}} \frac{g([F(r)]^{N-1})}{g([F(v)]^{N-1})} g'_s(1 - [F(v)]^N) N [F(v)]^{N-1} f(v) dv \\
&\leq g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) (v_0 - r) \\
&\quad + \int_r^{\bar{v}} \frac{[F(r)]^{N-1}}{[F(v)]^{N-1}} g'_s(1 - [F(v)]^N) N [F(v)]^{N-1} f(v) dv \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) (v_0 - r) + N [F(r)]^{N-1} \int_r^{\bar{v}} g'_s(1 - [F(v)]^N) dF(v) \\
&\leq g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) (v_0 - r) + N [F(r)]^{N-1} g'_s(1 - [F(r)]^N) (1 - F(r)) \\
&= g'_s(1 - [F(r)]^N) N [F(r)]^{N-1} f(r) \left[v_0 - \left(r - \frac{1 - F(r)}{f(r)} \right) \right]. \tag{33}
\end{aligned}$$

If g_s is strictly convex, the second inequality holds strictly.

Proof of Lemma 6

Inequality (22) is established in the following steps.

We first transform the integral in the inequality. Using substitution $t = F(v)$, the integral on the left-hand side (LHS) becomes $\int_{F(r)}^1 g'_s(1 - t^N) dt$. Further substitution $s = 1 - t^N$ transforms the integral into $\int_0^{1-[F(r)]^N} g'_s(s) \frac{ds}{N(1-s)^{(N-1)/N}}$. We thus need to show

$$\int_0^{1-[F(r)]^N} g'_s(s) \frac{[F(r)]^{N-1}}{(1-s)^{(N-1)/N}} ds \leq \int_{\{1-[F(r)]^N\}-N[F(r)]^{N-1}(1-F(r))}^{1-[F(r)]^N} g'_s(s) ds,$$

Equivalently, we can show

$$\int_0^{1-\alpha^N} g'_s(s) \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds \leq \int_{1-\alpha^N-N\alpha^{N-1}(1-\alpha)}^{1-\alpha^N} g'_s(s) ds, \tag{34}$$

where $\alpha = F(r)$. Let $a = 1 - \alpha^N - N\alpha^{N-1}(1 - \alpha)$ and $b = 1 - \alpha^N > a$. The LHS integral

has a density $\frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}}$ over $s \in [0, b]$, with a total weight:

$$\int_0^b \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds = N\alpha^{N-1}(1-\alpha).$$

The RHS has a density 1 over $s \in [a, b]$, with the same total weight:

$$b - a = N\alpha^{N-1}(1-\alpha).$$

Therefore, we have

$$\int_0^a \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds = \int_a^b ds - \int_a^b \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds.$$

Since $s \leq b = 1 - \alpha^N$, we have $1 - s \geq \alpha^N$. Raising to the power $(N-1)/N$:

$$(1-s)^{(N-1)/N} \geq \alpha^{N-1}.$$

We thus have

$$\begin{aligned} & LHS - RHS \\ &= \int_0^a g'_s(s) \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds + \int_a^b g'_s(s) \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds - \int_a^b g'_s(s) ds \\ &\leq g'_s(a) \int_0^a \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds + \int_a^b g'_s(s) \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} ds - \int_a^b g'_s(s) ds \\ &= g'_s(a) \int_a^b \left(1 - \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} \right) ds - \int_a^b g'_s(s) \left(1 - \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} \right) ds \\ &= \int_a^b (g'_s(a) - g'_s(s)) \left(1 - \frac{\alpha^{N-1}}{(1-s)^{(N-1)/N}} \right) ds \\ &\leq 0. \end{aligned}$$

which proves the inequality (34). Moreover, it is clear from the final inequality that (34) holds strictly if g_s is strictly convex.

Proof of Proposition 7

Fix any $r \in [v_0, \bar{v})$. We start with the case that buyers are risk neutral: $g_b(x) = x$. Let V_1 and V_2 denote the first and second highest valuations, respectively. In the SPA, bids are truthful. Hence, the seller's expected payoff is $E[\max\{r, V_2\} | v_1 \geq r]$ if $v_1 \geq r$ and v_0 otherwise. In the FPA, the risk neutral buyer with the highest valuation wins by bidding

his estimate of the maximum of the second-highest valuation and the reserve, assuming that his own valuation is the highest (Milgrom, 2004 Theorem 4.6). Hence, the seller's expected payoff is $E[E[\max\{r, V_2\}|V_1 = v_1]|v_1 \geq r]$ if $v_1 \geq r$ and v_0 otherwise. By the law of iterated expectations, the mean payoff is the same in each format. But the FPA payoff distribution is a mean preserving contraction of the SPA payoff distribution. A seller with (strictly) convex g_s is strongly risk averse and therefore (strictly) prefers the FPA payoff distribution.

Now take $g_b \in \mathcal{G}^*$, such that buyers are weakly risk averse. By Proposition 1, each participating type submits a weakly higher bid than when $g_b(x) = x$. If g_b is strictly star-shaped, the comparison is strict for almost every type. Hence, the distribution of FPA payoffs conditional on g_b first order stochastically dominates the distribution of FPA payoffs under risk neutrality, and the comparison is strict for strictly star-shaped g_b . As noted above, DU preferences respect monotonicity. A seller with convex g_s therefore prefers the FPA under g_b to the FPA with risk neutral buyers, and the FPA with risk neutral buyers to the SPA with risk neutral buyers. But the SPA with risk neutral buyers and the SPA under g_b coincide.

A.3 Entry Fee Results

This section contains the proofs of Propositions 3 and 5. The second builds on the first.

Fix (r, e) and let $\tilde{r} := v_{\dagger}(r, e)$ denote the associated participation threshold and $q_{\dagger} := F(v_{\dagger}(r, e))$ the quantile of the corresponding valuation. Let r_z denote the solution to

$$z \cdot e = g_b \left(F(\tilde{r})^{N-1} \right) (\tilde{r} - r_z)$$

for $z \in [0, 1]$. For the second-price auction, it is helpful to observe that

$$r_z = \tilde{r} - \alpha e z$$

where we define $\alpha := 1/g_b \left(F(\tilde{r})^{N-1} \right)$ to economize on notation.

We focus on the case that $e > 0$ and $v_{\dagger}(r, e) < \bar{v}$. If $e = 0$, the conclusion holds trivially. If $v_{\dagger}(r, e) = \bar{v}$, there is almost surely no entry under (r, e) or $(\tilde{r}, 0)$.

Proof of Propositions 3

First-price auction Let V_1 denote the highest valuation and $Q_1 = F(V_1)$ its quantile. For a given realization of (q_1, k) , the seller's payoff is

$$\pi_z(q_1, k) = \begin{cases} v_0 & \text{if } k = 0 \\ b(F^{-1}(q_1); r_z, ez) + kez & \text{if } k \geq 1 \end{cases} \quad (35)$$

Comparing $z = 1$ and $z = 0$, we have:

$$\begin{aligned}\pi_1(q_1, k) - \pi_0(q_1, k) &= \begin{cases} 0 & \text{if } k = 0 \\ b(F^{-1}(q_1); r, e) + ke - b(F^{-1}(q_1); \tilde{r}, 0) & \text{if } k \geq 1 \end{cases} \\ &= \begin{cases} 0 & \text{if } k = 0 \\ ke - \frac{e}{g_b(q_1^{N-1})} & \text{if } k \geq 1 \end{cases}\end{aligned}$$

where the second equality follows from (9).

Let Π_z denote the seller's payoff given z , a random variable. Then

$$E[\Pi_1 - \Pi_0] = eE\left[K - \frac{1}{g_b(Q_1^{N-1})} | K \geq 1\right] P(K \geq 1)$$

By the law of iterated expectations,

$$E[\Pi_1 - \Pi_0] = eE_{Q_1}\left[E[K | K \geq 1, Q_1 = q_1] - \frac{1}{g_b(q_1^{N-1})} | K \geq 1\right] P(K \geq 1)$$

Conditional on q_1 , the remaining $N - 1$ bidders enter with probability $(q_1 - q_{\dagger})/q_1 = 1 - \frac{q_{\dagger}}{q_1}$. Hence,

$$E[\Pi_1 - \Pi_0] = eE_{Q_1}\left[1 + (N - 1)\left(1 - \frac{q_{\dagger}}{q_1}\right) - \frac{1}{g_b(q_1^{N-1})} | K \geq 1\right] P(K \geq 1)$$

As q_1 is the first order statistic of N , the density of q_1 given $K \geq 1$ is $Nq^{N-1}/(1 - q_{\dagger}^N)$ over $[q_{\dagger}, 1]$. Moreover, $P(K \geq 1) = 1 - q_{\dagger}^N$. Therefore

$$\begin{aligned}E[\Pi_1 - \Pi_0] &= eN \int_{q_{\dagger}}^1 \left\{ N - (N - 1) \frac{q_{\dagger}}{u} - \frac{1}{g_b(u^{N-1})} \right\} u^{N-1} du \\ &= eN \left[\int_{q_{\dagger}}^1 \{Nu^{N-1} - (N - 1)q_{\dagger}u^{N-2}\} du - \int_{q_{\dagger}}^1 \frac{u^{N-1}}{g_b(u^{N-1})} du \right] \\ &= eN \left[(1 - q_{\dagger}^N) - q_{\dagger}(1 - q_{\dagger}^{N-1}) - \int_{q_{\dagger}}^1 \frac{u^{N-1}}{g_b(u^{N-1})} du \right] \\ &= eN \left[(1 - q_{\dagger}) - \int_{q_{\dagger}}^1 \frac{u^{N-1}}{g_b(u^{N-1})} du \right]\end{aligned}$$

The term in brackets is non-positive by star-shapedness of g_b . For a risk neutral seller, $U_s^{FPA}(r_z, ez) := E[\Pi_z]$ yields (10) in the main text.

Second-price auction Let V_1 and V_2 denote the first and second highest valuations, respectively. For a given realization of (v_1, v_2, k) , the seller's payoff is

$$\pi_z(v_1, v_2, k) = \begin{cases} v_0 & \text{if } k = 0 \\ \tilde{r} + (1 - \alpha)ez & \text{if } k = 1 \\ v_2 + kez & \text{if } k \geq 2 \end{cases} \quad (36)$$

where we note that $\alpha \geq 1$ and hence $1 - \alpha \leq 0$. Comparing $z = 1$ and $z = 0$, we have:

$$\pi_1(v_1, v_2, k) - \pi_0(v_1, v_2, k) = \begin{cases} 0 & \text{if } k = 0 \\ (1 - \alpha)e & \text{if } k = 1 \\ ke & \text{if } k \geq 2 \end{cases}$$

Let Π_z denote the seller's payoff given z , a random variable. Then

$$E[\Pi_1 - \Pi_0] = e \left\{ (1 - \alpha)p_1 + \sum_{k \geq 2} kp_k \right\} = e \left\{ -\alpha p_1 + \sum_{k \geq 1} kp_k \right\}$$

where $p_k = P(K = k)$. Since $K \sim \text{Binomial}(N, 1 - F(\tilde{r}))$, we have:

$$\begin{aligned} E[\Pi_1 - \Pi_0] &= e \left\{ -\alpha N(1 - F(\tilde{r}))F(\tilde{r})^{N-1} + N(1 - F(\tilde{r})) \right\} \\ &= eN(1 - F(\tilde{r})) \left\{ 1 - \alpha F(\tilde{r})^{N-1} \right\} \end{aligned} \quad (37)$$

By star-shapedness of g_b , the term in braces is non-positive. For a risk neutral seller, $U_s^{SPA}(r_z, ez) := E[\Pi_z]$ yields (11) in the main text.

Proof of Proposition 5

The proof extends the arguments of Proposition 3 above.

First-price auction For a given realization of q_1 define $\bar{\pi}_z(q_1) = E[\pi_z(Q_1, K) | Q_1 = q_1]$. Let $\bar{\Pi}_z$ denote the associated random variable. $\bar{\Pi}_1$ is trivially a mean preserving contraction of Π_1 . In other words, $\bar{\Pi}_1$ second order stochastically dominates Π_1 : $\bar{\Pi}_1 \geq_{SSD} \Pi_1$.

Observe that $\bar{\Pi}_0 = \Pi_0$, because Π_0 does not depend on K . To complete the proof, we now show that $\bar{\Pi}_0 \geq_{SSD} \bar{\Pi}_1$. This is enough for a risk averse seller (convex g_s) to prefer the reserve-only mechanism, $z = 0$. Because $\bar{\Pi}_0 | \{K = 0\} = \bar{\Pi}_1 | \{K = 0\}$, it suffices to show that $\bar{\Pi}_0 | \{K \geq 1\} \geq_{SSD} \bar{\Pi}_1 | \{K \geq 1\}$. We do so by showing that the integrated quantile function associated with $\bar{\Pi}_1 | \{K \geq 1\}$ lies below that of $\bar{\Pi}_0 | \{K \geq 1\}$.

Write $\tilde{\pi}_z(q_1) = E[\pi_z(Q_1, K) | Q_1 = q_1, K \geq 1]$ as the realization of $\bar{\Pi}_z | \{K \geq 1\}$ given q_1 and observe that $\tilde{\pi}_z(\cdot)$ is increasing on $[q_\dagger, 1]$. Furthermore, we know that

$$L(q) := P(Q_1 \leq q | K \geq 1) = \frac{N}{1 - q_\dagger^N} \int_{q_\dagger}^q s^{N-1} ds = \frac{q^N - q_\dagger^N}{1 - q_\dagger^N}$$

Let $\tilde{H}_z(\cdot)$ denote the distribution function for $\bar{\Pi}_z | \{K \geq 1\}$ and observe that:

$$\tilde{H}_z(\tilde{\pi}_z(L^{-1}(s))) = s$$

Because $\tilde{\pi}_z(\cdot)$ is monotonic, the s quantile of the payoff distribution is the payoff associated with the s quantile of Q_1 . Hence,

$$\int_0^p [\tilde{H}_1^{-1}(s) - \tilde{H}_0^{-1}(s)] ds = \int_0^p [\tilde{\pi}_1(L^{-1}(s)) - \tilde{\pi}_0(L^{-1}(s))] ds$$

Change of variables $L(q_1) = s \Rightarrow L'(q_1) dq_1 = ds$, and

$$\int_0^p [\tilde{H}_1^{-1}(s) - \tilde{H}_0^{-1}(s)] ds = \left(\frac{1}{1 - q_\dagger^N} \right) \int_{L^{-1}(0)}^{L^{-1}(p)} [\tilde{\pi}_1(q_1) - \tilde{\pi}_0(q_1)] N q_1^{N-1} dq_1$$

Fix $\tilde{q} := L^{-1}(p)$ for some $p \in [0, 1]$ and note $L^{-1}(0) = q_\dagger$. Then

$$\begin{aligned} \int_0^p [\tilde{H}_1^{-1}(s) - \tilde{H}_0^{-1}(s)] ds &= \left(\frac{N}{1 - q_\dagger^N} \right) \int_{q_\dagger}^{\tilde{q}} [\tilde{\pi}_1(q_1) - \tilde{\pi}_0(q_1)] q_1^{N-1} dq_1 \\ &= \left(\frac{eN}{1 - q_\dagger^N} \right) \int_{q_\dagger}^{\tilde{q}} \left(E[K | Q_1 = q_1, K \geq 1] - \frac{1}{g_b(q_1^{N-1})} \right) q_1^{N-1} dq_1 \\ &= \left(\frac{eN}{1 - q_\dagger^N} \right) \int_{q_\dagger}^{\tilde{q}} \left(1 + (N-1) \left(1 - \frac{q_\dagger}{q_1} \right) - \frac{1}{g_b(q_1^{N-1})} \right) q_1^{N-1} dq_1 \\ &= \left(\frac{eN}{1 - q_\dagger^N} \right) \left\{ \left((\tilde{q} - q_\dagger) \tilde{q}^{N-1} - \int_{q_\dagger}^{\tilde{q}} \frac{q_1^{N-1}}{g_b(q_1^{N-1})} dq_1 \right) \right\} \\ &\leq \left(\frac{eN}{1 - q_\dagger^N} \right) \left\{ \left((\tilde{q} - q_\dagger) - \int_{q_\dagger}^{\tilde{q}} \frac{q_1^{N-1}}{g_b(q_1^{N-1})} dq_1 \right) \right\} \end{aligned}$$

The RHS is non-positive by star-shapedness of g_b . As this holds for each $p \in [0, 1]$, the integrated quantile function associated with $\bar{\Pi}_1 | \{K \geq 1\}$ lies below that of $\bar{\Pi}_0 | \{K \geq 1\}$.

Second-price auction Let H_z denote the distribution of Π_z . From (37), we have that $E[\Pi_0] \geq E[\Pi_1]$. From (36), given $\alpha \geq 1$ and $e \geq 0$, we must have $H_0(t) \leq H_1(t)$ for all

$t \leq v_{\dagger}$ and $H_0(t) \geq H_1(t)$ for all $t \geq v_{\dagger}$. In other words, H_0 crosses H_1 once from below. Combining these observations, Π_0 second order stochastically dominates Π_1 . If g_s is convex, the seller therefore prefers Π_0 to Π_1 : again letting $\tilde{r} := r_0$, we have $U_s^{SPA}(\tilde{r}, 0) \geq U_s^{SPA}(r, e)$.

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